

Quality Improvement in the Modern Business Environment

The meaning of quality, its dimensions, and the methods used to improve it

Slides covered: **4-7, 19, 32**

Summary + key concepts

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1 The Meaning of Quality

slide 4

Quality is multi-dimensional. The text gives **the Eight Dimensions of Quality** that describe quality in most industrial and many business situations:

#	Dimension	Question it answers
1	Performance	Will the product do the intended job?
2	Reliability	How often does the product fail?
3	Durability	How long does the product last?
4	Serviceability	How easy is it to repair the product?
5	Aesthetics	What does the product look like?
6	Features	What does the product do?
7	Perceived Quality	What is the reputation of the company or its product?
8	Conformance to Standards	Is the product made exactly as the designer intended?

Additional Dimensions for Service Organizations

In service and transactional business organizations (banking and finance, health care, customer service), three more dimensions are added:

1. **Responsiveness** — How long did it take the service provider to reply to your request for service?
2. **Professionalism** — The knowledge and skills of the service provider; the competency of the organization to provide the required services.
3. **Attentiveness** — Customers want caring, personalized attention from their service providers.

Fitness for Use — the Traditional Definition

There are two general aspects of **fitness for use**:

① Quality of Design

The intentional differences in grade or quality level between products that serve the same function.

② Quality of Conformance

How well the product **conforms to the specifications required by the design**.

⚠ Note

This (quality = fitness for use) is the **traditional definition** of quality.

★ Modern Definition

“Quality is inversely proportional to variability.”

key relation

$$\text{Quality} \propto \frac{1}{\text{Variability}}$$

As **variability** in the important characteristics of a product **decreases**, the quality of the product **increases**. This is why reducing variability is the central goal of statistical quality control.

Three statistical methods are useful for quality improvement:

① **Statistical Process Control (SPC)**

- Control charts, plus other problem-solving tools.
- Useful in monitoring processes and reducing variability through **elimination of assignable causes**.
- An **on-line** technique (in-process procedure).

② **Designed Experiments (DOX)**

- Discovering the **key factors** that influence process performance.
- Process optimization.
- An **off-line** technique.

③ **Acceptance Sampling**

Inspection and classification of a sample of units selected at random, related to acceptance or rejection of the product.



Quick contrast

SPC = on-line (monitoring during production) · **DOX = off-line** (design / optimization).

Deming recommended the **Shewhart cycle** as a model to guide improvement. Its four steps — **Plan-Do-Check-Act** — are called the **PDCA cycle**. When the Check step is called *Study*, it becomes the **PDSA cycle**.

Step	Meaning
Plan	Propose a change in the system aimed at improvement.
Do	Carry out the change, usually on a small or pilot scale , to learn the results.
Check (Study)	Analyze the results to determine what has been learned about the changes.
Act	Adopt the change, or — if it was unsuccessful — abandon it.

⚠ **Key point**

The process is almost always **iterative**, and may require several cycles to solve complex problems.



Quick Review

- **8 dimensions of quality** + **3 service dimensions** (Responsiveness, Professionalism, Attentiveness).
- **Fitness for use** = Quality of design + Quality of conformance → *traditional definition*.
- **Modern definition:** quality is **inversely** proportional to variability.
- **3 statistical methods:** SPC (on-line) · DOX (off-line) · Acceptance sampling.
- **Improvement cycle:** PDCA / PDSA — iterative.